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Still emerging: understanding Generative AI use in Higher Education

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Introduction and methods: As an emerging technology, Generative Artificial Intelligence (GAI) tools are currently subject to constant re-inventions, and they may be employed in widely different ways by users. This study sought to develop understanding pertaining to the adoption of GAI tools on the basis of a cross-sectional survey conducted within a specific institutional context. It explored the following research question: "How do staff and students in an academic setting perceive their own responses to the specific emerging technology of Large Language Models?".
Results: The study found that appropriate uses of GAI tools in the research setting were not straightforwardly obvious for participants. Variation was identified in both the use of GAI tools and in expertise on the part of staff, so that it would have been difficult for many staff participants to appreciate educationally appropriate uses of GAI tools and the extent to which students were manifesting those uses. While student respondents were more positive than staff about likely gains to be had from making use of these tools, and considered that facilitating conditions to support their use of GAI tools were in place, the main use that came consistently into view for students was to use GAI tools to improve language-usage and for text production.
Discussion: The findings suggest that more is needed than to frame the challenges posed by GAI tools for Higher Education across the world as a technical challenge for learning design or for training that addresses the AI literacy of students. The affordances of GAI tools play a key in determining their effective use. These affordances need to be imagined more adequately by both students and staff if this technology is to serve educational purposes effectively, with this imagination on the part of staff also needing to account for understanding of the ways that students themselves imagine the affordances of GAI tools.

KEYWORDS
affordances, assessment, emerging technology, generative artificial intelligence, large language models

Introduction

Generative Artificial Intelligence (GAI) has recently gained significant popularity across Higher Education, with a widespread adoption of specific GAI tools that are referred to as Large Language Models (LLMs). LLMs such as ChatGPT and Claude have the capacity to produce meaningful blocks of text in response to even brief

prompts (Rudolph et al., 2023). Such tools are able to produce similar cultural artefacts to those that are created by staff and students as an integral aspect of their professional work or their academic studies. As a result, there is scope for LLMs to reshape practices in Higher Education on a widespread basis. Indeed, a tendency is evident in the research literature to position LLMs in terms of either their potential positive impact on education or the chaos they will bring with them. Many members of staff within higher education are concerned about the potential for GAI tools, including those focused on the production of other artefacts than text, to destabilise higher education (Carrigan, 2024). Attention, though, has also concentrated on the possibility for GAI to transform and enhance learning and teaching (Adıgüzel et al., 2023; Nazir and Wang, 2023).

Such a dichotomous framing of the impact of technology reflects a broader tendency for utopian and dystopian framings to emerge in the wake of nascent technologies (Carrigan and Fatsis, 2021). As commentators such as Boyd (2014) have long pointed out, these perspectives are mirror images of each other, in the sense they assume the power of the technology to enact change while underplaying, or even entirely obviating, the significance of human agency to bringing about these effects in real contexts. As Boyd (2014, p. 313) puts it: "Utopian rhetoric assumes that when a particular technology is broadly adopted it will transform society in magnificent ways, while dystopian visions focus on all of the terrible things that will happen because of the widespread adoption of a particular technology that ruins everything."

There is a need, then, to develop understanding of what staff and students are doing with GAI as the technology is adopted across Higher Education, as well as to develop understanding of the extent to which such adoption has been effective (or not) in educational terms. This constitutes a challenging task to undertake, though, given that the digital tools involved may be employed in widely different ways by users and they are currently subject to constant re-inventions. Indeed, Holmes and Tuomi (2022) argued that GAI can appropriately be characterised as an emerging technology. Emerging technologies refer to technological developments which are recent, ongoing and significant in their potential social and economic implications. Examples of emerging technologies include developments such as quantum computing and augmented reality.

This article reports on a study that sought to develop understanding pertaining to the adoption of GAI tools within Higher Education. The perceptions of staff and students are particularly important in appreciating GAI tool use in this setting. Indeed, a focus on perceptions assists in viewing GAI tools as sociotechnical objects, as technical capacities interact with social processes in given contexts. The study thus addressed on the following research question: "How do staff and students in an academic setting perceive their own responses to the specific emerging technology of LLMs?" This research question lends itself to an empirical investigation within a specific institutional context, not so much to shore up practice within that setting, but as a space in which to develop a detailed understanding of GAI tool adoption. The study was conducted within a School of a research-intensive university. An institutional policy on the use of GAI was already in existence, with this policy requiring that students cite any text or paraphrasing of text from GAI tools that is incorporated into assessed work. It takes time, though, for institutionally-endorsed practices and infrastructure to develop, even while the technology itself shifts. It will be important, then, to interpret findings on the perceptions of a given set of users in light of a close understanding of the nature of the emerging technology in play. Prior to outlining the research design that underpins this study, we thus first explore existing perspectives on the nature of LLMs. A related paper reports on a further stage to the research that sought, additionally, to explore ways in which staff sought to exercise agency as teachers in response to student use of GAI tools (Kahn et al. 2025).

Literature review: understanding generative AI

Education as a broad sector has particularly emerged as a key area of focus for practical application of emerging technologies (Criollo-C et al., 2023): major technological advancements have "led society almost always towards making ever-increasing demands on education" (Leoste et al., 2021, p. 110). In turn, education is striving to deeply engage with the societal implications of emerging technologies and to adapt to the evolving social needs they create (Bearman and Ajjawi, 2023; Leoste et al., 2021). It should be recognised that the category "emerging technologies" itself can be used performatively to generate interest in a particular technology and create a sense of inevitability around it. The history of educational technology is filled with promises of radical change that were never realised, often put forward by those with a commercial interest in the technology in question (Weller, 2020). However, a sociological awareness of the role which classifying a set of digital tools as an emerging technology plays in commercialisation processes does not mean we should discount the potential significance of the technology in question.

While the definition of the term "emerging" is contentious, an emerging technology is typically characterised as fundamentally new in either methods (how it works) or functions (what it achieves), whether they are born out of revolutionary innovations or evolutionary advancements (Rotolo et al., 2015; Veletsianos, 2010). Technologies evolve by adapting to new domains, appearing radically novel in these contexts (Rotolo et al., 2015). The adaptation to new domains feeds back into the design and subsequent development of the technology itself. An equally significant criterion for characterising a given domain of digital tools as an emerging technology, though, is their ability to create substantial transformations, impacting, for instance, organisations, knowledge production, and technological regimes (Veletsianos, 2010). As a result of this novelty and impact, emerging technologies are further characterised by experiences of uncertainty and ambiguity on the part of users.

What kind of emerging technology is ChatGPT, as the most popular of a series of publicly available models at the furthest frontier of technological capacities? There is a technical answer to this question which would identify it as a LLM constituted through a training process utilising a set of training data, before post-training activities work to optimise and channel the model's response to maximise its utility for end users. Naveed

et al. (2025) indicated that the technical capabilities of LLMs have seen extensive advances in recent years, with a wide range of new applications opened up as a result. These applications include those that pertain to learning, teaching and assessment within universities (Adıgüzel et al., 2023; Nazir and Wang, 2023). ChatGPT, however, is not simply encountered by students as a technical object. It is also experienced as a cloud-hosted software package that is characterised by a particular set of affordances. Davis (2020, p. 25) explains that technologies "don't make people do things but instead, push, pull, enable and constrain", that is, they have affordances. Technologies encourage and discourage, request and demand, and they influence actions for their users in ways that are socially situated. For example, at the time of authoring this article, ChatGPT offers the advice "What can I help with?" and the prompt "Ask anything". The firms which operate platforms and software actively steer the relationship which users have with their software, to establish particular kinds of relationships between the user and the system (Carrigan and Fatsis 2021). The fact GAI tools are general purpose leaves users influenced by the discourse surrounding them, more than for tools which have specific and obvious uses, with scope for this discourse to shift in unpredictable ways.

As a result, user engagement is always filtered through imagined affordances rather than being an immediate encounter with functionalities and constraints that are objectively part of the software. Indeed, the relational character of affordances complicates *objectivity*, in the sense they don't straight forwardly inhere in the software itself but rather manifest themselves through the interaction (Archer, 2000). If the user does not imagine the software can be used for a particular purpose, they will not use the technology in such a way. The perceptions and understanding of users are thus integral to appreciating technology use. Nagy and Neff (2015, p. 1) argue that "Imagined affordances emerge between users' perceptions, attitudes, and expectations; between the materiality and functionality of technologies; and between the intentions and perceptions of designers". It is particularly interesting to consider this within universities where educators might take on a parallel role to that of the designer, in terms of scaffolding the uptake of emerging technologies through the implicit and explicit expectations which students are confronted with concerning their use. The notion of imagined affordance offered by Nagy and Neff (2015, p. 5) integrates "the various ways of theorizing affordance to create a concept flexible and robust enough for the complex emerging socio-technical relationships in social life".

If one is to understand the nature of GAI as it is currently playing out within Higher Education, it is thus important to engage with the perspectives of educators and students as they engage with these novel digital tools. Understanding the categories through which people make sense of emerging technologies, as well as the work they do they undertake in using them is critical (Couldry, 2012). As a technology emerges it is not a straightforward matter to understand its implications for education and for institutions. This is why attending to the affordances of an emerging technology needs to go hand-in-hand with considering how those affordances are being framed, interpreted and understood by those for whom use will be obligated, expected or simply desirable within university settings. This is particularly relevant to GAI tools, given that their uses are so open-ended.

Methodology

The research design was formulated as a case study that drew on survey research method (Clark, et al., 2021), with two surveys for use within the case setting of a School that formed part of a research-intensive university in the UK and that covered a set of Social Science disciplines Such a design incorporates systematic methods for collecting large amounts of data to describe, compare, or explain the nature of certain conditions, attitudes, and behaviours (Cohen et al., 2017; Gray, 2014), enabling us to develop explanations pertaining to the phenomenon in view. By employing such a research design, we aimed not only to illustrate the occurrence of a phenomenon in a specific context —and concerning a particular population at a particular moment—but also to offer explanations that will be of value to others in similar situations who face comparable research or practical concerns (Andres, 2012). Yin (2018) refers to such an approach to external validity as analytic generalisation.

We designed two original online questionnaires: one for students and the other for academic staff members (see Supplementary Appendix for all questionnaire items that contribute to validated scales). To develop these instruments, we reviewed existing scales related to technology acceptance and use, with these primarily based on the Technology Acceptance Model (TAM) (Davis et al., 1989) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003). UTAUT represented a direct extension of TAM, with these two models providing Likert scales pertaining to technology adoption that have been rigorously developed and extensively used in educational research over the last few decades. Davis et al. (1989) argued that it is important to understand why users accept or reject technology, demonstrating the way in which factors such as the perceived usefulness, the perceived ease of use, one's attitude towards use, and one's behavioural intention to use all influence whether or not someone will actually make use of the technology. Given the specific focus of our study on GAI, we also adapted elements from more recent models, such as the Artificially Intelligent Device Use Acceptance (AIDUA) model (Gursoy et al., 2019) and the Integrated AI Acceptance-Avoidance Model (IAAAM) (Cao et al., 2021). The questionnaires were thus predominantly constructed with close-ended questions, primarily using Likert scales, to measure a set of perceptions on the part of staff and students about their use of GAI tools.

By adapting some items and introducing novel ones, the resulting questionnaires facilitated both a descriptive approach—aimed at measuring the nature and scale of occurrences—and an analytical one, which enabled us to engage with and build upon established theories in the field (Gray, 2014), an approach that is aligned to recent studies of a similar kind (Baig and Yadegaridehkordi, 2025). While the notion of "acceptance" as interpreted in these specific models does not fully acknowledge the extent to which different users can imagine the affordances of GAI tools in varied ways, or the uncertainties that might accompany

GAI tool adoption, this approach nonetheless provided a baseline from which to explore such complexities. The survey additionally included novel scales that sought to explore perceptions that could not be conceptualised in light of the above-indicated models, with attention given to affectivity in the use of GAI tools and to perceptions of specific uses to which others were using GAI tools to support academic study: given the influence of these dimensions on the scope to imagine the affordances of GAI tools. The study does not seek to measure affordances directly, but takes the perceptions of staff and students as indicators of how the affordances of GAI tools are imagined.

The quality of an instrument's measurement is determined in part by analysing the characteristics and content of its items (Flores and Terán, 2022). To ensure this, a content validation process was conducted based on expert judgment (Cabero and Barroso, 2013). Experts were selected according to a competence coefficient (K) of 0.8 or higher, which was calculated considering their self-assessed familiarisation and argumentation capacity on the subject under study (Marín-González et al., 2021). A panel of seven experts evaluated the sufficiency, relevance, clarity, and coherence of the items in both surveys, following the criteria proposed by Escobar and Cuervo (2008). The degree of agreement among the judges (Aiken's V values above 0.8) indicates that the designed items and dimensions were appropriate for measuring the constructs of the selected theoretical model. Finally, the resulting questionnaire for academic staff comprised 11 dimensions and included a total of 48 items, with response options on a 7-point Likert scale. The questionnaire for students, on the other hand, covered 9 dimensions and included a total of 40 items.

Ethics approval was granted by the University of Manchester Research Ethics Committee (review reference 2024-19365-34383), with personal data protection regulations considered as an integral aspect of the application.

Data collection

Questionnaires were designed to be self-administered (Clark et al., 2021). They were simultaneously disseminated via the online platform Qualtrics between April and June 2024. Participants were invited through social media, e-newsletters, email invites, and (for students) the distribution of flyers in a student centre. While the capacities and perceptions of staff and students in the research site have shifted since then, LLMs themselves have been shifting rapidly as well since they first came to global prominence. For instance, a user who briefly experimented with ChatGPT3.5 might assume they have a working understanding of the capacities of the current GPT5 model, but there are significant differences, nonetheless in the functionality of these two models. The perceptions of users are by their nature impressionistic, risking a conflation of personal factors (i.e., becoming more adept at using the model) and technological factors (i.e., the model actually becoming more capable). Even so we suggest they should be taken seriously as a data point.

We collected 45 complete responses from a population of 318 academics, including Lecturers, Senior Tutors, and Researchers (with two additional teaching assistants also responding). Further responses were included in the dataset, comprising partial survey responses from another 26 staff members. The relatively low response rate represents a limitation on the study, but it is likely that a self-selection bias occurred in favour of those with an existing interest in GAI tools. This observation is supported by the reflection that several members of staff indicated to members of the research team that they had not completed the survey (or fully completed it) because they found that they had insufficient knowledge to answer some questions. The responses were predominantly from highly experienced staff, with 64% having over ten years of experience in Higher Education. In terms of gender distribution, the respondents consisted of 42% female and 51% male, with 7% indicating "Prefer not to say". Additionally, we gathered 86 complete responses from students, although some questions received up to 116 responses—from a population of around 3,400. Of the total student respondents, 66% were from postgraduate taught programmes, with the remaining respondents coming from undergraduate programmes. While this represents a relatively modest sample size, the approach taken to data analysis (as outlined below) ensures that the findings are based on a rigorous validation process.

Data analysis

We conducted an internal consistency analysis to evaluate whether items within single measures adapted from existing instruments consistently measured the same construct. This analysis provided us with empirical evidence to support scales that were deemed conceptually sound. Specifically, we utilised Cronbach's alpha to assess the internal consistency of scales and an indicator reliability test to evaluate the extent to which each item contributes to the scale, with a minimum level of 0.70 considered acceptable in both cases). Scales for the following constructs were effectively validated for the staff survey: Effort Expectancy, Facilitating Conditions, Performance Expectancy, Negative Attitude, Perception of Students' Use of GAI and Intention of Use. For the student survey, meanwhile, the following constructs were validated: Facilitating Conditions, Performance Expectancy, Negative Attitude and Use of GAI for Academic Purposes. Full details for these scales are reported in the Supplementary Material. A set of further scales did not achieve successful validation and are thus excluded from the reporting of findings. In addition, we provide evidence from certain items that were designed to yield predominantly factual responses and that were thus not intended to contribute to scales. According to DeVellis and Thorne (2021), the development of a scale is not anticipated when a survey aims to gather simple and readily understandable information from respondents, particularly when it is closely related to behavioural indicators.

In this article, we present the means and standard deviations of univariate scales and pertinent items. Validated scales represent the average value for all items contributing to the specified scale, and they are shown in their original 1–7 Likert scale for consistency in their interpretation. Furthermore, when comparisons are made between identical scales and items from both populations, we also provide the significances from *t*-tests.

Findings

It is first of all helpful to report on the main items that were designed to yield predominantly factual responses. We draw on such responses in so far as they assist in explaining the overall nature of GAI tool adoption in the research setting. Responses indicated that 63% of staff respondents had used at least one GAI tool at some point in the 6 months to May 2024, although only 14% of respondents had taken out a paid subscription. By contrast, students reported a higher uptake (21%) of paid subscriptions to GAI tools compared to staff.

Perceived ease of use

The dimension of the survey that related to the perception that use of GAI tools would be free from undue effort was validated for the staff survey but not for the student survey, with responses indicated in Figure 1. Relatively few respondents endorsed the notion that extensive effort was required to make use of GAI. This reflects the way in which the use of LLMs such as ChatGPT and Claude straightforwardly rely on the existing capacity of users to employ natural language. What is not clear from this scale, though, is the extent to which staff respondents felt they were able to employ GAI tools to easily complete a *diverse* range of academic tasks.

Conditions that facilitate the adoption of GAI tools

Both surveys returned a validated scale for the extent to which respondents perceived facilitating conditions to be in place to support their use of GAI tools, with the results presented in Figure 2. A *t*-test (with *p*-value=0.001) indicated that students perceived to a greater extent than staff that facilitating conditions to support their use of GAI tools were in place. The mean for student responses was 4.94 compared to 4.15 for staff. While it was thus the case that a larger proportion of student respondents compared to staff respondents indicated that they had access to subscription-only tools, both sets of respondents would have had access to the fundamental technology itself through free-to-access versions on their own hardware.

Extent to which use of GAI tools enhances academic performance

Scales on both surveys were successfully validated for the extent to which GAI tools were perceived by respondents to enhance their performance in completing relevant academic tasks, with the results presented in Table 1. The impact of GAI tools on performance depends on many influences, including which tools are employed, the manner in which the tools are used, institutional or professional endorsements for usage of tools in given settings, and so on. It is worth observing that the distribution of responses on the student survey was skewed to the right (*p*-values <0.05 when running a D'Agostino test).

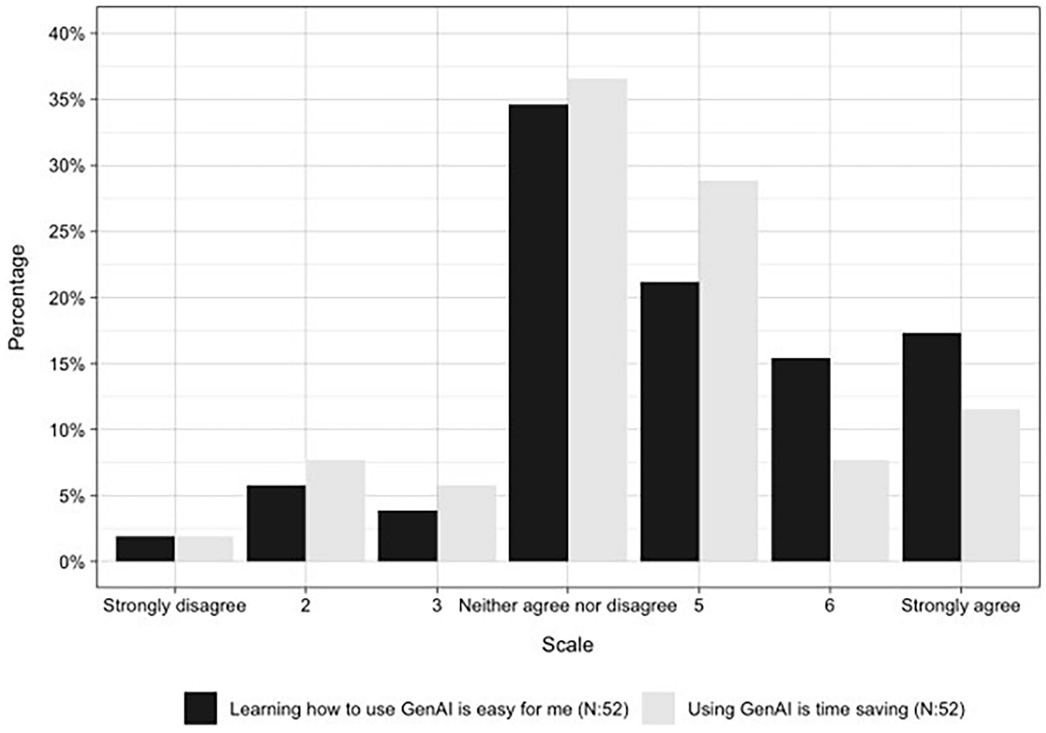


FIGURE 1
Extent to which GAI tools were perceived as easy to use by staff respondents. Responses are shown on a 1–7 Likert scale, where 1 = strongly disagree and 7 = strongly agree. Scale values represent the mean of validated items contributing to the Effort Expectancy scale.

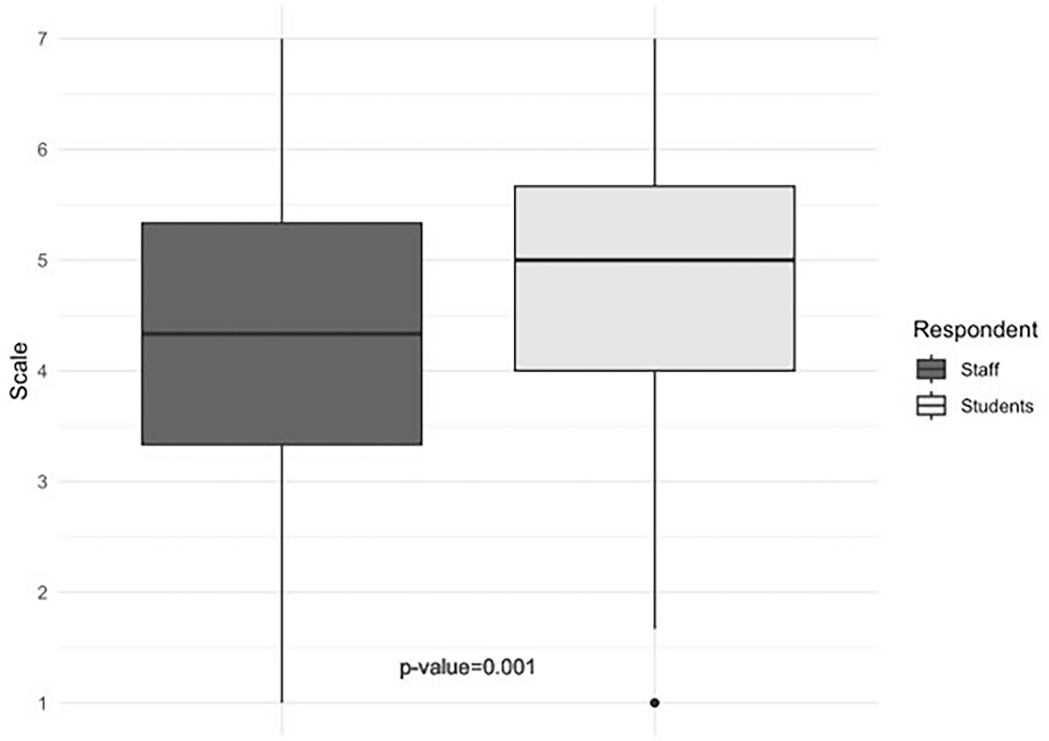


FIGURE 2
Comparison for staff and student respondents of the perceptions of facilitating conditions to support the use of GAI tools, based on the Facilitating Conditions scale for staff and the Facilitating Conditions scale for students.

TABLE 1 Extent to which the use of GAI is believed to help in performance.

Item (*n*)	Index		On a scale of 1 to 7, where 1 is "strongly disagree" and 7 is "strongly agree" (%)						
	x̄	SD	1	2	3	4	5	6	7
Performance expectancy –staff	3.9	1.6							
The use of GenAI could ease my research workload (52)			11.5	7.7	11.5	30.8	19.2	11.5	7.7
The use of GenAI could ease my teaching workload (51)			11.8	13.7	15.7	17.6	23.5	7.8	9.8
Using GenAI increases my productivity (49)			12.2	6.1	10.2	38.8	18.4	6.1	8.2
Performance expectancy – students	5	1.6							
The use of GenAI can significantly reduce students workload (116)			6.9	4.3	6.9	17.2	20.7	19	25
Using GenAI increases my productivity (116)			4.3	6	4.3	19	17.2	20.7	28.4
The use of GenAI helps me learn about my subject (115)			9.6	5.2	3.5	11.3	19.1	22.6	28.7
My academic performance is likely to improve by using GenAI (115)			7	6.1	8.7	20	22.6	13	22.6

While a direct comparison between the responses of staff and students is not appropriate given the different items that were used across the two surveys, the skew indicates a positive response on the part of respondents on this dimension of the student survey. The mean of 5 on this scale for the student responses would suggest that specific use cases were typically in view for the student respondents. The data presented in Table 1, though, would suggest that suitable affordances of GAI tools in relation to easing workloads and increasing productivity are not straightforwardly in view for many staff respondents.

Negativity in attitudes towards student use of GAI tools

Figure 3 presents the validated scale on negativity in one's attitudes towards the use of GAI for academic purposes. It is perhaps not surprising that student respondents showed less negativity in their attitudes towards the use of GAI than staff respondents. The mean for staff on this scale was significantly greater than the mean for students (4.93 vs. 4.10, as confirmed by a *t*-test with a p-value < 0.001). Chan and Hu (2023), indeed, found that the students in their survey were generally positive

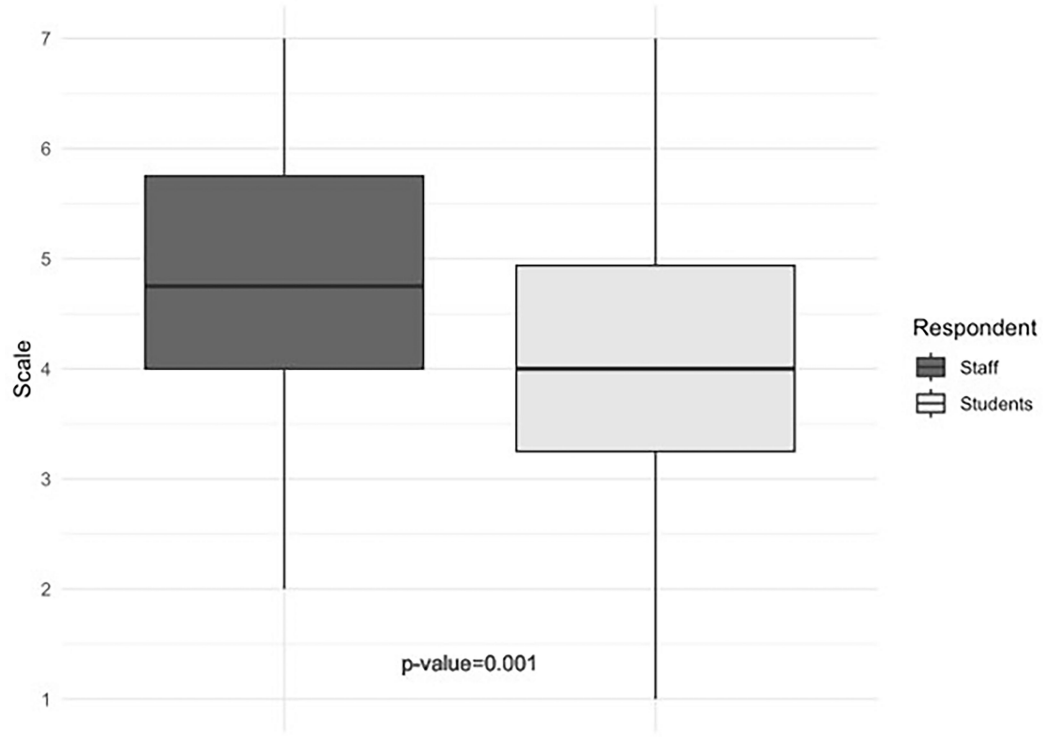


FIGURE 3
Comparison for staff and student respondents on negativity in the attitudes towards students' use of GenAI for academic purposes, based on the Negative Attitude scale for staff and the Negative Attitude scale for students.

towards the use of GAI in their studies. The attitudes of staff towards student use of GAI tools depend on their appreciation of both the affordances of tools and on what they have seen in terms of student work that they suspect has been affected by a use of GAI tools. Staff, though, only possess a limited view on the actual uses to which students put GAI tools, given students' uses of such tools are not generally visible to staff. It is, for instance, not straightforward for members of staff to tell whether or not a student has used a GAI tool in completing an academic task when only the final text is in view (Elkhatat et al., 2023). In addition, there is the possibility that the views of staff may be formed by their understanding of the functionality of GAI tools at an earlier stage of tool development.

Perceptions of students' use of GAI tools for academic purposes

The same set of items loaded onto the validated scales for both staff and students, namely: looking for answer for given tasks and assignments; generating any type of written assignment; and translation or improving the grammar of writing. Note that these three items connect to student use of LLMs to support writing tasks, with support for language-usage and the production of text the two main uses that are clearly in view. In itself the use of an LLM by a student to generate written text for use in an assessment represents a fundamental shift in what a technology can achieve from what was openly possible a few years ago. Using an LLM for the immediate purpose of text production, though, seems relatively passive (at least in educational terms) in comparison to the use, say, of LLMs to improve one's own understanding or to analyse research data. It may well be more difficult to imagine affordances that stray beyond a focus on text production, given that this is the most immediate functionality of current LLMs.

Students perceived that other students use GAI for academic purposes to a significantly lesser extent than staff members perceive students use GAI (see Figure 4). The mean for this scale for staff was 5.23, compared to 3.11 for students (with a *t*-test confirmed at $p = 0.000$). It is possible that some students may have underplayed their use of GAI tools in responding to the survey, but these results, nonetheless, highlight the limited knowledge that staff possess in relation to student use of GAI tools. In this case, the gap will be partly down to the prevalence of student use of GAI tools while learning independently of staff (Hockings et al., 2018); and thus to the restricted nature of the social relations that staff maintain with students as they engage in academic study.

Intention to use GAI tools

Staff respondents were asked whether they were likely to make use of GAI in their research, teaching and other work-related tasks, with a validated scale resulting around intention to use GAI tools. It was clear that some members of staff were intending to make use of GAI in their work, while others were not (see Figure 5, below). If members of staff are disengaged from using GAI in their own working practice, though, it is

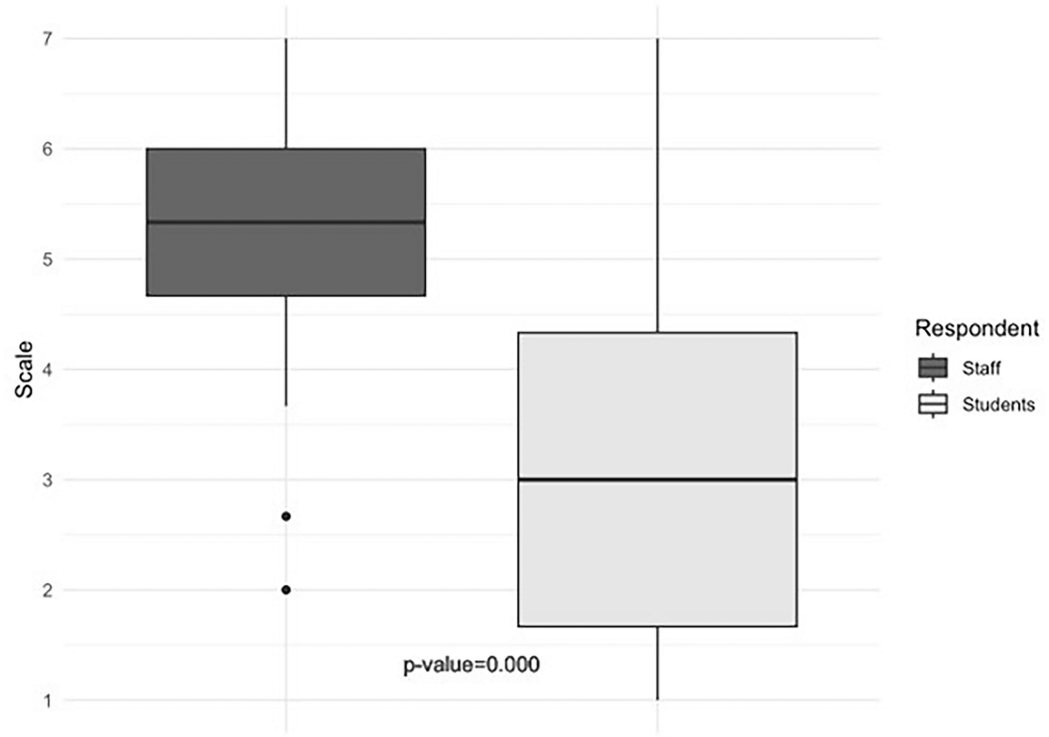


FIGURE 4
Comparison for staff and students on the perceptions about students' use of GAI tools for academic purposes, based on the Perception of Students' Use of GAI scale for staff and the Use of GAI for Academic Purposes scale for students.

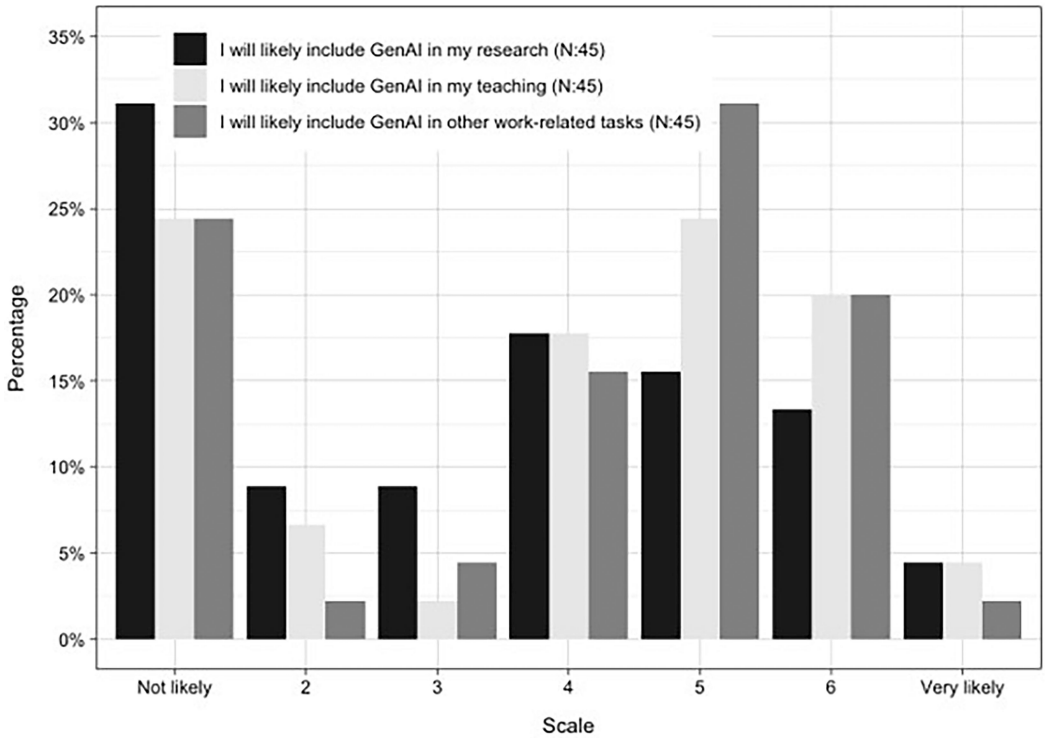


FIGURE 5
Intention of use in the professional practice of staff respondents. Responses are shown on a 1–7 Likert scale, where 1 = not likely and 7 = very likely. Scale values represent the mean of validated items contributing to the Intention of Use scale for staff.

difficult to see how they will be in an effective position to scaffold educationally-appropriate uses of these tools on the part of students.

Discussion

Emerging technologies such as GAI will tend to produce uncertainty and ambiguity for users within applied settings. They provide affordances for entirely new practices, such as the real time generation of multimedia in a classroom setting or new ways of undertaking familiar practices, such as using a chatbot to answer an essay question. There is nothing in these developments, though, which determines a response by users, since the affordances will be perceived differently by users based on any number of personal or institutional factors. How GAI will be evaluated within an educational organization depends on how the affordances are regarded by users. It follows from this that perceptions of GAI will tend to be varied across the potential users within an organization. This is complicated further by the tendency for debates concerning emerging technologies to become polarized between utopian and dystopian perspectives (Boyd, 2014; Carrigan, 2025), as already noted. This variety and polarization could be seen clearly in our findings, with a wide range of views held by staff in the school on issues such as ease of use, increases in productivity, the likely impact on student creativity and criticality and the extent to which staff were assisting each other with GAI tools; and with staff and students perceiving facilitating conditions to support the use of GAI tools to be in place to differing extents.

Uncertainty, though, was seen to emerge not simply in relation to ways in which respondents perceived their own use of GAI tools, but also in relation to other users. In an educational context, the perceptions that staff hold about student activity represent a fundamental aspect of GAI tools when they are conceived as sociotechnical objects. The findings highlighted differing levels of negativity towards use of GAI tools on the part of staff and students, and limitations to the knowledge that staff possess in relation to student use of these tools. Ambiguity thus occurs not simply in relation to the varied (and changing) nature of the GAI tools themselves, and thus to the uses to which the tools can be put, but also in terms of the social relations that staff and students maintain with each other. This fills out the perspective offered by Rotolo et al. (2015) that an emerging technology can develop in varied ways because different social groups perceive the technology in diverging ways. There is a relational dimension stemming from the nature of education that is closely relevant to the way in which educators seek to scaffold the ways that students make use of GAI tools. The issue here is not just how a user imagines a set of affordances for a tool, but how educators imagine students imagining the affordances of GAI tools. Given this it is important to remember that core considerations of the emerging dimension to technology, such as novelty, impact and uncertainty apply to the social dimension of technology as well as to its technical aspects.

It is also important to consider why staff and students perceived the tools in different ways. It was striking that while 63% of staff respondents had used at least one GAI tool in the six months prior to completing the survey, only 14% of respondents had used a paid subscription. If we assume that staff who were engaged with GAI, whether through a positive or a negative lens, were more likely to respond to the survey, then it should be a surprise this subscription rate was so low. While the specific details vary from month to month in a competitive market, free versions of the software tend to be rate limited and (until recently) restricted to older versions of the model. This means that users relying on a free-to-access version only have practical experiences of the affordances GAI in a restricted form, while the 37% of staff had never used a GAI tool had no such practical experience at the time. Mollick (2024) argued that practical experimentation is needed to develop a working understanding of GAI tools. For this reason a subscription can be taken as a meaningful proxy for sustained engagement, suggesting that a substantial majority of our respondents were working with affordances that were substantively imagined to varying degrees. It is notable that reported student engagement with subscriptions was slightly higher at 21%, in spite of the fact that students could be expected, as a group, to have less disposable income available for subscriptions than staff, and certainly no access to research budgets to subsidize these costs.

A greater level of sustained engagement with the real affordances and the limitations of GAI tools on the part of students could be a factor in explaining the different evaluations between staff and students. Students were more positive than staff about the likely gains from using GAI tools to support their studies and reduce their workloads, and about the extent to which they had appropriate support. While it would be unwarranted to assume this *must* reflect greater engagement with the software, as opposed to, say, a more positive evaluation of digital technologies as a category, there are prima facie grounds for believing the level of sustained engagement constitutes a causal factor amongst others. Additional explanations of the differences between staff and student responses will, though, be possible. For instance, Orlikowski and Gash (1994) argued that the way in which different users make use of new technologies will depend on the frames of reference by which they make sense of the technology, with potential for those frames to differ between groups of users in any given organisational setting. One would quite naturally expect students to be more concerned than staff about the time involved in producing assessed work, with scope for incongruence between staff and students to result.

The findings, furthermore, provide the basis for a critique of the Technology Acceptance Model (TAM) in light of consideration of ways in which GAI constitutes an emerging technology. Existing studies have pointed out that TAM downplays the cultural dimensions of technology use, and that the term "acceptance" frames the use of technology in a passive manner (Stockman, 2017). While also drawing attention to the relevance of such critique for emerging technologies in particular, the study highlights how use of GAI tools in educational settings is caught up in a relational dynamic that was not directly appreciated in the critique of the TAM by Stockman (2017). Much of the challenge over student use of GAI tools pertains to the restricted nature of the social relations that staff maintain with students.

It is thus important to stress the agency of both educators and students. If we reject the deterministic assumption that a technology will dictate the uses which are made of it, such that its social effects are an expression of an intrinsic nature, then questions of (contextualised) agency will always be important, as explored further in Kahn et al. (2025). The decisions which individuals make "on the ground", as well as the settings within which and in relation to which they make those decisions, will be crucial to understand dynamics of integration and application. Mollick (2024), indeed, pointed out that GAI tools are a general-purpose technology which by their nature requires substantial exploration and experimentation across different domains. However, this possibility relies on those within organisations having the time and autonomy required to experiment with their use—exactly what the imperative to automate using these newfound affordances threatens to undercut (Carrigan, 2024).

Conclusions

This study offers insights into how a specific emerging technology is seen to play out within a specific educational setting. It clarifies understanding of ways in which different users of that technology responded to its emergence, and the roles that they played in relation to each other. The study found that appropriate uses of GAI tools in the research setting were not straightforwardly obvious for participants, but subject to ambiguity. For this reason, we agree with Bearman and Ajjawi (2023, p. 1160) who argued, "Rather than seeking to explain 'blackboxes', we argue that a pedagogy for an AI-mediated world involves learning to work with opaque, partial and ambiguous situations, which reflects the entangled relationships between people and technologies". It is precisely this entanglement which our research design has sought to unpick, particularly the converging or diverging orientations to and attitudes about GAI tools. The findings suggests that more is needed than to frame AI literacy as a linear phenomenon in which problems are ameliorated in proportion to the level of training provided.

It was clear in this study that staff possessed limited knowledge about ways in which students were making use of GAI tools. Indeed, variation was identified in both use and expertise on the part of staff, so that it would have been difficult for many staff participants to appreciate educationally appropriate uses of GAI tools and the extent to which students were manifesting those uses. While student respondents were more positive than staff about likely gains to be had from making use of these tools, and considered (at least, in comparison to staff respondents) that facilitating conditions to support their use of GAI tools were in place, the main use that came consistently into view for students through these survey was to support language-usage and text production. Perhaps this is not surprising if these tasks represent those that are immediately in view to users, with many members of staff in the research setting unlikely to be in a position to open up wider uses for their students. User acceptance as it is currently understood thus emerges as a thin concept that is not able to portray the extent to which an effective educational use of GAI tools depends on inter-relationships between staff and students.

To the extent we start from the assumption that our students are inclined to use GAI tools to circumvent assessment integrity, we start from a position of distrust (Carrigan, 2024). Obviously, this does not mean we should deny that students can and will use these tools in problematic ways to circumvent assessment practices, only that we should not treat the construction of this as a problem in a straightforward and naive manner. This will mean that learning and assessment changes to account for the affordances of GAI tools, rather than seeking to reinscribe the existing system in a Canute-like effort to repel the diffusion of a technology which has already disseminated. From the perspective of Dewey and Rogers (2012) problems always make sense relative to publics, which is to say that we should not treat a problem as something which reality presents to us an objective impediment but rather as a phenomenon which needs to be understood relationally (Carrigan and Fatsis, 2021).

This invites us to reconstruct the educational challenges raised by GAI tools in terms of the teacher/student and corresponding peer relationships which comprise the setting in which reflexivity is being exercised. This is not just a matter of rational deliberation but also the modes of feeling which are entailed by the emerging technology and its reception, given how the disruptive expectations attached to them inevitably carry and real and imagined implications for the relationships within which we are embedded and which give our lives meaning (Papacharissi, 2015). The psychological complexity of what Carrigan (2024, p. 9) calls "the great assessment panic" needs to be incorporated into our analysis, resisting the temptation to reduce this to a technical challenge for learning design or the AI literacy of students. There is a much more complex process underway in which relationships within educational settings are being recalibrated through the causal influence of GAI tools, raising the question of how this process might instead by reflexively steered in a manner leading to improved teacher/student relations.

In considering avenues for future research, it is helpful to consider some of the study's limitations. One could clearly undertake a study with a larger (and thus potentially more balanced) sample to address issues that might arise from under-coverage of relevant sub-groups within the population (e.g., to avoid over surveying those with high levels of interest in LLMs) (Gray, 2014). It would be particularly valuable, though, to pursue qualitative research that provided a rich description of the perspectives of students on use of LLMs within their academic studies; something that the current study does not offer. Given the subjective nature of imagined affordances, this would provide an opportunity to consider how issues such as purpose, trust and affectivity influence the uses to which students put LLMs. It would also enable a keener understanding of ways in which student use of LLMs in their academic studies can be understood in relational terms.

The affordances of GAI need to be imagined effectively by both students and staff if the technology is to serve educational purposes effectively, with this imagination on the part of staff also needing to account for understanding of the ways that students themselves imagine the affordances of GAI tools. Imagination on the part of staff of ways in which students

conceive their use of technology is an integral aspect of what it means to be an educator. As staff and students in universities continue to adjust their expectations and practices to a technology that is still emerging, it is essential that a fuller appreciation of the complexities in play around the educational use of GAI tools come into view.

Data availability statement

The datasets presented in this article are not readily available because ethics approval was not secured for release of datasets. Requests to access the datasets should be directed to PK, peter.kahn@manchester.ac.uk.

Ethics statement

The studies involving humans were approved by University Research Ethics Committee, University of Manchester. The studies were conducted in accordance with the local legislation and institutional requirements. The ethics committee/institutional review board waived the requirement of written informed consent for participation from the participants or the participants' legal guardians/next of kin because (for the completion of the anonymous questionnaire only): implied consent provided on the basis of an explicit statement was considered acceptable for this purpose - following standard practice for anonymous questionnaires on non-sensitive issues.

Author contributions

PK: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. MC: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Writing – original draft, Writing – review & editing. IW: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. RL: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Writing – review & editing. PS: Conceptualization, Funding acquisition, Investigation, Methodology, Writing – review & editing. AA: Conceptualization, Data curation, Investigation, Methodology, Writing – review & editing. LM: Conceptualization, Funding acquisition, Writing – review & editing. FS: Conceptualization, Data curation, Methodology, Writing – review & editing.

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Supplementary material

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